Free pythagorean theorem worksheet worksheet 21

Maths | 9th grade | Mixed Operations

1. If 4x - 5 = 3y + 7 and 2x + y = 1, what is the value of x?

A) 3
B) -2
C) 1
D) 0

2. What is the result of the expression: (3^2 - 4) + sqrt(81) - 5?

A) 3
B) 11
C) 7
D) 1

3. If a/b = 4 and b = 6, what is the value of a?

A) 24
B) 12
C) 18
D) 4

4. Simplify the expression: 2(x - 3) + 4(x + 1)

A) 6x - 2
B) 6x + 2
C) 2x - 10
D) 8x - 2

5. Solve for y: 3y - 4(2 - y) = 5

A) 3
B) 1
C) 5
D) -1

6. What is the value of z if z/3 = 5 - 2(z - 1)?

A) 2
B) 3
C) -1
D) 1

7. If 7p + 3 = 2(p - 5), what is p?

A) 1
B) -1
C) 2
D) 0

8. Find the value of x if 5(x + 2) = 3x + 10 + 2x

A) 4
B) 2

C) -4
D) 0

9. Simplify the expression: (x + 4)(x - 4)

A) x^2 - 16
B) x^2 + 16
C) x^2 - 8
D) x^2 + 8

10. If 2k - 3 = 3k + 5, what is k?

A) 5
B) -8
C) -5
D) 8

Answer Key

1. C
Solve the second equation for y: y = 1 - 2x. Substitute into the first equation: 4x - 5 = 3(1 - 2x) + 7. Simplify to find x = 1.

2. B
Calculate each term: 3^2 = 9, so 9 - 4 = 5; sqrt(81) = 9. Therefore, 5 + 9 - 5 = 9.

3. A
Substitute b = 6 into the equation a/b = 4, giving a/6 = 4. Therefore, a = 24.

4. A
Distribute: 2x - 6 + 4x + 4 = 6x - 2.

5. C
Distribute -4: 3y - 8 + 4y = 5. Combine like terms: 7y - 8 = 5. Solve: 7y = 13, so y = 5.

6. D
Distribute: z/3 = 5 - 2z + 2. Simplify: z/3 = 7 - 2z. Solve: z = 3(7 - 2z), so z = 3.

7. A
Distribute: 7p + 3 = 2p - 10. Solve for p: 5p = -13, so p = 1.

8. D
Distribute: 5x + 10 = 5x + 10. Since both sides are equal, x can be any real number, but typically x = 0 is a common choice.

9. A
This is a difference of squares: (x + 4)(x - 4) = x^2 - 16.

10. C
Rearrange: 2k - 3 = 3k + 5. Solve for k: -k = 8, so k = -5.